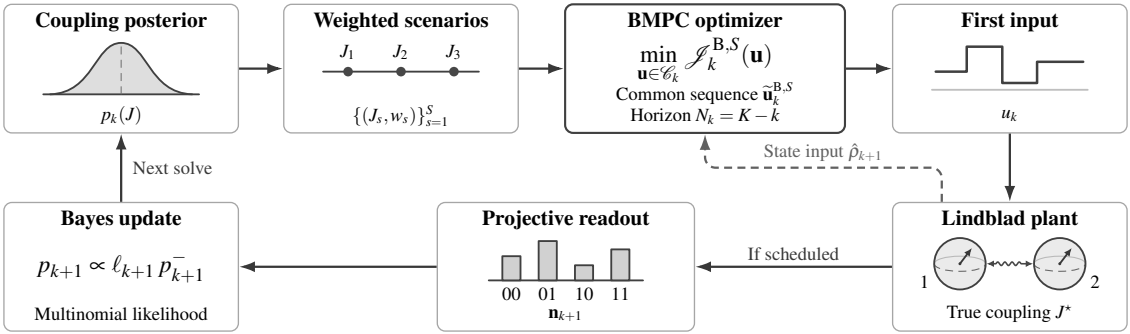

Coupling posterior
$p_k(J)$
Weighted scenarios
J_1 J_2 J_3
$\{(J_s, w_s)\}_{s=1}^{S}$
BMPC optimizer
$\min_{\mathbf{u}\in\mathscr{C}_k} \mathscr{J}_k^{\mathrm{B},S}(\mathbf{u})$
Common sequence $\widetilde{\mathbf{u}}_k^{\mathrm{B},S}$
Horizon $N_k = K - k$
First input
u_k
State input $\hat{\rho}_{k+1}$
Lindblad plant
1
2
True coupling $J^\star$
If scheduled
Projective readout
00 01 10 11
$\mathbf{n}_{k+1}$
Bayes update
$p_{k+1} \propto \ell_{k+1}\, p_{k+1}^{-}$
Multinomial likelihood
Next solve